

WAVE EQUATION

$$\boxed{\frac{\partial^2 \varphi}{\partial t^2} = c^2 \frac{\partial^2 \varphi}{\partial x^2}}$$

$$\begin{aligned} x &\in [0, L] \\ t &\in [0, \infty[\end{aligned}$$

posto $u = \varphi$
 $v = \frac{\partial \varphi}{\partial t}$

allora:

$$\begin{cases} \frac{\partial u}{\partial t} = v \\ \frac{\partial v}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \end{cases}$$

ovvero

$$\frac{\partial}{\partial t} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & I \\ c^2 \frac{\partial^2}{\partial x^2} & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

Le condizioni iniziali sono nel vettore $[u, v]^T$

$$u(x, 0) = u_0(x) = \varphi_0(x)$$

$$v(x, 0) = v_0(x) = \frac{\partial \varphi_0}{\partial t}$$

Le condizioni al contorno sono, ad esempio:

$$u(x_B, t) = 0$$

$$v(x_B, t) = 0$$

$$\left. \begin{aligned} u(x_B, t) &= 0 \\ v(x_B, t) &= 0 \end{aligned} \right\} \begin{aligned} \varphi(0, t) &= \varphi(L, t) = 0 \\ \frac{\partial \varphi}{\partial t}(0, t) &= \frac{\partial \varphi}{\partial t}(L, t) = 0 \end{aligned}$$